

Inherent optical properties of Jerlov water types

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The diffuse attenuation coefficient $K_d(\lambda)$ was first expressed in terms of the inherent optical properties (IOPs) of water according to well-established empirical bio-optical models. Boltzmann simulated annealing was then used to find the best sets of IOPs to fit $K_d(\lambda)$ spectra to the reference spectra $K_d^0(\lambda)$ that define the Jerlov water types. Absorption $a(\lambda)$ and scattering $b(\lambda)$ coefficients were thus obtained for all Jerlov water types over the wavelength range 300–700 nm. The chlorophyll concentrations corresponding to the Jerlov water types were also obtained via bio-optical models. The result is a self-consistent set of spectral IOPs, chlorophyll concentrations, and Jerlov water types useful for a variety of underwater optical communications and remote sensing applications. © 2015 Optical Society of America

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1. INTRODUCTION

In the early days of optical oceanography, commercial instruments were not available for routine *in situ* measurement of inherent optical properties (IOPs; the absorption and scattering coefficients in particular). However, it was easy to measure profiles of downwelling plane irradiance $E_d(z, \lambda)$, from which the downwelling diffuse attenuation coefficient $K_d(z, \lambda) = [-1/E_d(z, \lambda)]d[E_d(z, \lambda)]/dz$ could be determined. This apparent optical property (AOP) gives a convenient measure of light penetration into the upper ocean. $K_d(z, \lambda)$ varies systematically with wavelength over a wide range of water types from very clear to very turbid, and it is rather insensitive to external environmental conditions such as solar zenith angle variations during typical K_d measurements. $K_d(z, \lambda)$ is thus considered a quasi-inherent optical property [1]. Models have been developed for determination of $K_d(z, \lambda)$ at any wavelength from a measurement at one wavelength [2], and $K_d(\lambda = 490 \text{ nm})$ is a standard product obtained from ocean-color remote sensors such as SeaWiFS, MODIS, and VIIRS.

In 1951, Jerlov obtained transmittance data from near-surface water clarity measurements made during the 1948 Norwegian circumnavigation expedition [3], the account of which constitutes an amazing read for any oceanographer. Transmittance at wavelength λ from the surface to depth z , $T(z, \lambda)$, is related to K_d via $T(z, \lambda) = \exp[-K_d(\lambda)z]$ (assuming that K_d is constant with depth, or using a depth-averaged K_d value). Based on the transmittance, i.e., on the spectral shape of $K_d(\lambda)$, Jerlov initially categorized waters into three distinct oceanic types (numbered I, II, and III) and five coastal

types (1C, 3C, 5C, 7C, and 9C) as shown in Figs. 1 and 2. He later added two additional subdivisions for oceanic type I water [3,4]. The Jerlov water types became a commonly used and convenient one-parameter classification scheme for describing water clarity for essentially all natural bodies of water [5].

A rough correspondence between the Jerlov water type and the chlorophyll concentration for some Case 1 waters can be made [6], but data on the IOPs corresponding to the Jerlov water types are fragmentary.

The use of K_d and Jerlov water types as descriptors of water optical properties fell into disfavor by many optical oceanographers when commercial instruments (e.g., the WETLabs ac-9 and bb-9, and the HOBILabs Hydrosat-6) for direct measurement of absorption, attenuation, and backscatter became widely available. Nevertheless, the Jerlov classification scheme still is useful, especially in applications where estimates of performance in typical water conditions are required. Such applications are related to underwater visibility [7] and communications [8,9], and they would be even more useful if the IOPs corresponding to each Jerlov type were available.

This paper therefore presents a method for determining IOPs for all Jerlov water types. We also describe an approach for reconstructing biological constituent concentrations from measurements of $K_d(\lambda)$. Jerlov's original $K_d(\lambda)$ spectra (hereafter referred to as reference values, denoted $K_d^0(\lambda)$), are combined with well-established empirical bio-optical models that relate AOPs and IOPs, thus providing a reference frame for determining $a(\lambda)$ and $b(\lambda)$ from $K_d(\lambda)$. Because these IOPs are modeled in terms of the concentrations of biological

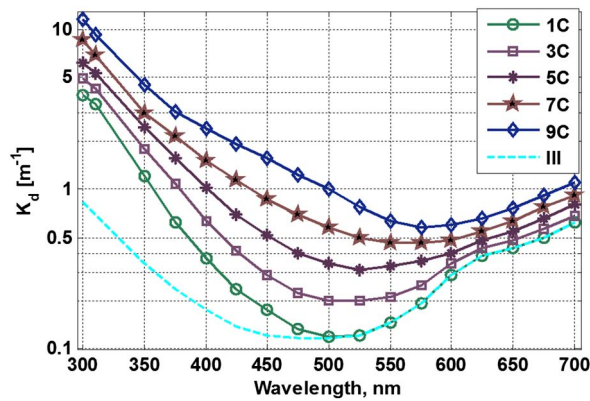


Fig. 1. Diffuse attenuation coefficients $K_d^0(\lambda)$ for Jerlov coastal water types 1C through 9C [3,4]. The Jerlov type III (oceanic) curve is shown for comparison.

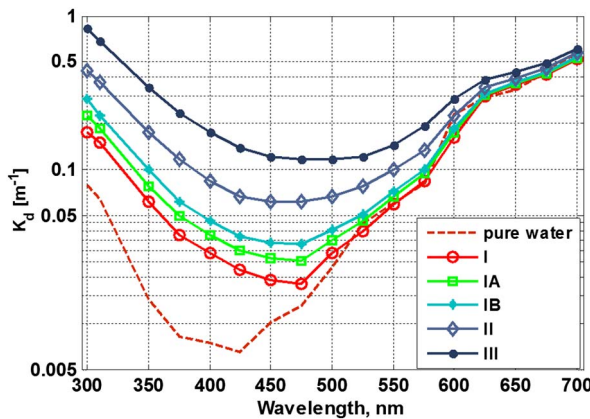


Fig. 2. Diffuse attenuation coefficients $K_d^0(\lambda)$ of Jerlov oceanic water types I through III [3,4] and the absorption spectrum $a(\lambda)$ of pure water [19–21].

constituents, the resulting absorption and scattering spectra also relate the Jerlov water types to measurable quantities of biological in-water constituents. This procedure provides a tool for studying the biological constituents of ocean waters based on easily measured or remotely sensed $K_d(\lambda)$ spectra.

The next section first reviews $K_d^0(\lambda)$ reference data. The ways in which $K_d(\lambda)$ is parameterized by IOPs, and in which the IOPs are, in turn, parameterized by the water constituents, are then presented. Section 3 then describes the Boltzmann simulated annealing (BSA) inversion algorithm. Section 4 uses HydroLight simulations to show the consistency of the retrieved IOPs and the $K_d^0(\lambda)$ spectra. Section 5 discusses the results and their implications for the oceanographic community. Table 1 shows the nomenclature used throughout.

2. DATA AND PARAMETERIZATIONS

This section further describes Jerlov's $K_d^0(\lambda)$ data and then the parameterizations of $K_d(\lambda)$ and the IOPs, all of which form the inputs to the inversion algorithm described in Section 3.

Table 1. Nomenclature of Abbreviations Used

Symbol	Definition	Units
$a(\lambda)$	Absorption coefficient	m^{-1}
$a_w, a_{\text{chl}}, a_{\text{cdom}}$	Absorption coefficients of pure water, chlorophyll, and CDOM, respectively	m^{-1}
$a_{\text{cdom}}^o, a_{\text{cdom}}^c$	CDOM absorption coefficients of ocean and coastal waters, respectively	m^{-1}
B_s, B_l	Concentrations of small and large particles in water, respectively	g m^{-3}
$b(\lambda)$	Scattering coefficient	m^{-1}
b_w, b_p	Scattering coefficients of pure water and particulates, respectively	m^{-1}
Chl	Chlorophyll concentration	mg m^{-3}
$K_d(\lambda)$	Diffuse attenuation coefficient	m^{-1}
$K_d^0(\lambda)$	Diffuse attenuation coefficient of original Jerlov data (i.e., reference values)	m^{-1}
M	Scaling coefficient of CDOM absorption, Eq. (7)	—
$\mathbb{C}$	Cost function used in the BSA optimization	—
α	Exponent of CDOM absorption, Eq. (7)	nm^{-1}
δ	Nonparametric contribution to cost function, Eq. (2)	—
η	Fraction of molecular scattering in total scattering, Eq. (3)	—
μ_ω	Average cosine of solar rays just beneath the water surface	—

A. Reference Data

The original Jerlov measurements of water transmission [3] were obtained for 15 to 17 spectral bands, typically at depths of 25 m for the ultraviolet (UV) portion of the spectrum and 50 m for the visible portion. These measurements were made mostly in the southern Pacific Ocean and the coastal regions of Norwegian fjords. The colored-glass filters used for the measurements had transmission bandwidths of 30 to 40 nm. Measurements in the UV and visible ranges were obtained in separate runs, utilizing instruments designed specifically for either UV or visible spectral ranges.

All measurements for oceanic water types (I through III) were conducted at a near-zenith Sun elevation angle of 75°. Measurements of coastal water transmission were typically performed at a Sun elevation angle of 45°, mostly in the Gullmarfjord area of Norway. For all cases, “calm” sea conditions were reported. K_d^0 values for coastal water types 3C, 5C, 7C, and 9C at 300 and 310 nm were obtained by extrapolating from the Jerlov data at longer wavelengths.

B. Parametrization of K_d

Two noncontradictory approaches can be used to describe the relationships among total absorption, scattering, constituent concentrations, and diffuse attenuation coefficients. The first approach is based on the works of Gordon [1], Morel and Maritorena [10], Morel and Loisel [11], and Bricaud *et al.* [12]. This approach was implemented for computation of IOPs for Case 1 waters in the HydroLight radiative transfer numerical model [13,14], which produces good K_d inversion results [15]. The second approach was developed mostly by Haltrin [16], using Kopelevich's equations [17] and other data.

Because the Gordon–Morel–Bricaud (GMB) approach provides a way to parameterize molecular versus particulate scattering, and because of computational efficiency considerations, the GMB method is used here to model total absorption and to relate water IOPs to AOPs. The approach developed by Haltrin and Kopelevich considers scattering as the sum of scattering caused by small and large particles. It does not explicitly depend on whether the water is Case 1 or Case 2 [16] and is thus more general than the GMB approach, which was developed for Case 1 waters only. Because of these reasons, the Haltrin–Kopelevich approach is used here to characterize multicomponent ocean water scattering.

GMB starts by introducing two scattering phase functions, one for molecules and one for particles. The total scattering phase function is then expressed as a weighted sum of the two [11]:

$$\beta(\phi) = \eta\beta_w(\phi) + (1 - \eta)\beta_p(\phi). \quad (1)$$

Here, $\beta(\phi)$, $\beta_w(\phi)$ and $\beta_p(\phi)$ are the total, molecular, and particle scattering phase functions, respectively. The shape of the $\beta_w(\phi)$ phase function can be found in [11]. The $\beta_p(\phi)$ function is the averaged Petzold phase function obtained by averaging the coastal, harbor, and ocean phase functions measured by Petzold [18]. The corresponding phase function average cosine g is 0.924.

The scattering phase function is thus separated into two parts that describe contributions from molecular scattering and highly forward-peaked particle scattering. The parameter η , the fraction of total scattering attributed to molecular scattering, is defined as

$$\eta = \frac{b_w}{b_w + b_p}, \quad (2)$$

where b_w is the coefficient of molecular scattering of pure water and b_p is the total particle scattering coefficient.

The diffuse attenuation coefficient for the depth at which the irradiance is 10% of the surface value can be written in terms of IOPs as [11,19]

$$K_d = \frac{a}{\mu_\omega} \sqrt{1 + \frac{b}{a} G(\mu_\omega, 2.3, \eta)}, \quad (3)$$

where $G(\mu_\omega, 2.3, \eta) = (0.451 + 2.584\eta)\mu_\omega - (0.205 + 0.521\eta)$.

The quantity μ_ω is the (average) cosine of solar rays just beneath the water surface. Equation (3) accounts for contributions from both small and large scattering particles as well as for the solar angle above the horizon. For Jerlov water types I through III (oceanic), $\mu_\omega = 0.98$ (Sun elevation angle of 75°). For water types 1C through 9C (coastal), $\mu_\omega = 0.85$ (Sun elevation of 45°).

Because most of the Jerlov measurements were taken at depths of 25–50 m, optical depth $\zeta \approx 2.3$ is an appropriate value to use here. The relation between the diffuse attenuation coefficient and the IOPs for Jerlov waters can then be written as

$$K_d = \frac{a}{\mu_\omega} \sqrt{1 + \frac{b\mu_\omega}{a} [(0.451 + 2.584\eta) - \frac{1}{\mu_\omega} (0.205 + 0.521\eta)]}, \quad (4)$$

where the coefficient η is computed using Eq. (2) as an average fraction of the molecular scattering over all wavelengths.

C. Parametrization of IOPs

Jerlov's oceanic water types (I through III) refer to Case 1 waters, i.e., to waters mostly free of inorganic particles and extra colored dissolved organic matter (CDOM) from terrigenous sources. Jerlov's coastal water types (1C through 9C) are generally Case 2. These waters can result from both biological processes and also a variety of nonbiological processes such as terrestrial runoff of CDOM and minerals and sediment suspension. Therefore, to better parameterize the whole range of Jerlov waters in a consistent manner, we adopt a model that (a) contains terms accounting for both molecular and particulate scatter and (b) preserves the independence of large and small particle concentrations. In addition, we express water absorption in terms of the concentrations of the most relevant components, namely chlorophyll and CDOM. In other words, this work combines the GMB model for absorption with the description of multicomponent ocean water scattering suggested by Haltrin [20].

The absorption coefficient of ocean water is written as

$$a(\lambda) = a_w(\lambda) + a_{\text{chl}}(\lambda) + a_{\text{cdom}}(\lambda). \quad (5)$$

The first term is the absorption spectrum of pure water. This spectrum is produced by averaging the data of Pope and Fry [21], Buiteveld *et al.* [22], and Pegau *et al.* [23] (Fig. 2). The second term is chlorophyll-related absorption, which depends on chlorophyll concentration Chl in the manner developed by Bricaud *et al.* [12] and used by Morel and Maritorena [10] and Stramski *et al.* [24]:

$$a_{\text{chl}}(\lambda) = A(\lambda)[Chl]^{E(\lambda)}. \quad (6)$$

This work adopts a case of medium UV absorption by chlorophyll. Values of $A(\lambda)$ and $E(\lambda)$ for this case, as given in [14], are shown in Fig. 3. The chlorophyll concentration, in mg m^{-3} , is treated as one of the optimization parameters during the inversion procedure (details below).

CDOM absorption is also expressed in terms of chlorophyll concentration [14]:

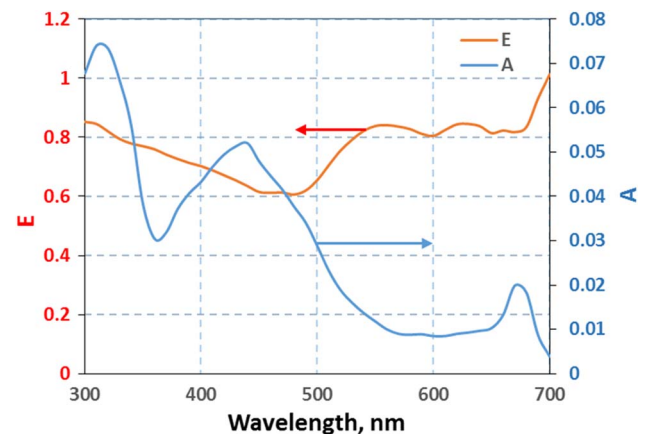


Fig. 3. Wavelength dependence of parameters A and E for the case of medium UV absorption by chlorophyll as defined in [12]. Arrows indicate the vertical axis of each chart.

$$a_{\text{cdom}}(\lambda) = a_{\text{chl}}(440)M \exp[-\alpha(\lambda - 440)]. \quad (7)$$

In the equation above, $a_{\text{chl}}(440)$ is chlorophyll absorption at 440 nm, with units of m^{-1} . The parameters M and α are fitting parameters that are optimized during the BSA procedure for each Jerlov water type. The initial values of M and α are set to 0.2 and 0.016, respectively, as per [14].

In the course of this work, an attempt was made to introduce an additional term into Eq. (5) to represent wavelength-independent absorption due to inorganic particulates. This term was treated as an independent parameter to be determined during the optimization. Interestingly, the term was routinely minimized by the BSA algorithm to a value below the standard deviation of the parameter, even for the most turbid Jerlov 9C water. The implication is that absorption, as is implicit in the Jerlov classification via the strong dependence of K_d on the absorption coefficient, can be modeled mostly by chlorophyll and CDOM regardless of whether the water is Case 1 or Case 2.

The optical model developed by Kopelevich [17] and later improved by Haltrin [20] describes the dependence of the total scattering coefficient $b(\lambda)$ on in-water organic constituents for a wide range of sea waters. Having two different size classes of scattering particles provides additional IOP generality in our parameterizations. Following [20], the total scattering attenuation coefficient b can be written as

$$b(\lambda) = b_w(\lambda) + b_p(\lambda), \quad (8)$$

where

$$b_w(\lambda) = 5.83 \cdot 10^{-3} \left(\frac{400}{\lambda} \right)^{4.322}, \quad (8a)$$

$$b_p(\lambda) = B_s b_s(\lambda) + B_l b_l(\lambda), \quad (8b)$$

$$b_s(\lambda) = 1.513 \left(\frac{400}{\lambda} \right)^{1.7}, \quad (8c)$$

$$b_l(\lambda) = 0.3411 \left(\frac{400}{\lambda} \right)^{0.3}. \quad (8d)$$

Here, wavelength is in nanometers, b_w describes the molecular scattering of pure water, b_s represents the total scattering coefficient of small particles, and b_l represents the total scattering coefficient of large particles. The coefficients B_s and B_l give the corresponding concentrations of small and large particles, in g m^{-3} .

3. INVERSION ALGORITHM

We now turn our attention to the inversion algorithm, which employs BSA.

A. BSA: The Cost Function

In general terms, the reconstruction of $a(\lambda)$ and $b(\lambda)$ from $K_d(\lambda)$ follows the methodology described in [25]. First, the spectrum of the diffuse attenuation coefficient is expressed in terms of spectra of the absorption and scattering coefficients as described above in Section 2.B. The latter are in turn parameterized in terms of the concentrations of relevant biological and inorganic constituents $P_1, P_2, \dots$, as in Section 2.B. Conceptually, then,

$$K_d = f(a(\lambda, P^a, P^a, \dots), b(\lambda, P^b, P^b, \dots), P_1, P_2, \dots). \quad (9)$$

Note that $f(a, b, P)$ does not explicitly depend on wavelength. The parameters P^a, P^b , and P are then adjusted so as to minimize the cost function,

$$\mathbb{C} = \sum_i \left| \frac{K_d(\lambda_i, P^a, P^b, P) - K_d^0(\lambda_i)}{K_d^0(\lambda_i)} \right| + \sum_j |\delta_j|, \quad (10)$$

for all wavelengths at the same time. Here, $P^a = [P_1^a, P_2^a, \dots]$, $P^b = [P_1^b, P_2^b, \dots]$, and $P = [P_1, P_2, \dots]$. The term K_d^0 is the reference spectrum of diffuse attenuation coefficients for a particular Jerlov water type. The symbol δ_j represents other contributors to the cost function—for example, the weighted difference between optimized and measured values of absorption or scattering at some particular wavelength. The absolute value is chosen instead of the usual quadratic form, because the latter is more sensitive to the high-absorption regions (i.e., the short- and long-wavelength wings) of the K_d spectra at the expense of the more practically significant intermediate blue-green range.

The cost function of Eq. (12), expressed in terms of the constituents of absorption and scattering, is highly nonlinear and may have multiple local minima. Therefore, BSA was selected as an optimization technique [26]. The BSA technique does not guarantee an absolute minimum of the cost function, but it can find an optimum solution, i.e., a set of underlying parameters that corresponds to a lowest local minimum, without the need for the excessive computational resources to examine every possible solution. Parameter boundary conditions and constraints can be imposed to reduce the sampling space and BSA computation time by assuring that only physically meaningful values of the parameters are explored.

The inverse problem for Jerlov waters—i.e., the reconstruction of $a(\lambda)$, $b(\lambda)$, and the concentrations of in-water constituents solely from $K_d(\lambda)$ —is, for several reasons, an ill-posed problem. First, inversion is limited by the information content of the original Jerlov data, that is, the number of wavelength sampling points and the accuracy of the $K_d(\lambda)$ measurements, particularly at short wavelengths. Combined with high absorption and scattering for coastal water types, contributions from key constituents such as chlorophyll and CDOM can couple and become difficult to discern. In addition, Jerlov's measurements do not describe in sufficient detail a number of environmental parameters important for analytical modeling of the diffuse attenuation coefficient, in particular the scattering phase function, and factors such as possible shading by the vessel. Therefore, the absorption and scattering coefficients reconstructed here for Jerlov waters represent an optimum approximation of water scattering and absorption that provides the best match of IOPs to AOPs for the original Jerlov data, consistent with the chosen bio-optical models. The assumptions of no diffuse sky radiance contribution and an average scattering cosine $g > 0.8$ [11] are inherent in this approach. The BSA procedure was performed on the assumption that Sun zenith angle did not change during the measurements.

B. BSA: Details

BSA examines randomly selected sets of optimization parameters with respect to the value of the cost function given in

Eq. (10). An examined set is selected as a current optimum if it generates a smaller cost function than the previous set. In addition, a set may be selected as a current optimum even if the generated cost function is not minimal, in this case with a probability defined by a probabilistic acceptance function inspired by the Boltzman distribution function of classical statistical mechanics. This probabilistic function is proportional to $\exp(-C/T)$, where C is the absolute value of the cost function and T is a parameter analogous to temperature. This function defines the probability of accepting a nonoptimal solution produced by the current set of optimization parameters. With each newly tested parameter set, T is reduced according to a specified “cooling schedule.” At the termination of the model run, the parameter set with the minimum cost function value is declared an optimum solution. The BSA algorithm used in this work is described in detail in [26].

Generally, the accuracy of the BSA solution depends on the number of tries the algorithm attempts for the given number of optimization parameters, the initial guess of the parameter values, and the specified parameter constraints. The smaller the difference between the initially guessed parameter set and the parameter set that eventually produces the true (global) minimum of the cost function, the more accurate the solution. The smaller the search range of the parameters, the more accurate is the solution reached within the allotted computational time. In other words, algorithm efficiency can be increased by incorporating *a priori* knowledge of the approximate solution location.

To improve algorithm efficiency and remove the dependence of results on the initial parameter-set guess, the BSA procedure of [26] was modified for this work. After the BSA returned a set of parameters representing some minimum of the cost function, the algorithm was run again, this time with the new initial-guess parameter set C_i defined as the cost-weighted average of the previously returned coefficients:

$$C_i = \frac{\sum_k c_i^k \varepsilon_k}{\sum_k \varepsilon_k}, \quad (11)$$

$$\varepsilon_k = \frac{1}{C_k}, \quad (11a)$$

where r_i^0 is the initial range of parameter i . Typically, the BSA algorithm was run 20–30 times for each data set (i.e., each Jerlov water type), and the parameters corresponding to the absolute minimum of the cost function in all runs were reported as the BSA optimization result. To estimate the error introduced by the algorithm itself, this BSA optimization (again, consisting of 20–30 BSA runs) was repeated 3–7 times. Thus, 3–7 values were obtained for each parameter, and these values were used to determine averages and standard deviations. The BSA optimization settings are summarized in Table 2.

As described in the previous section, the BSA nominally optimizes five parameters (Chl , α , M , B_s , and B_l), using as its criterion the difference between the computed values of $K_d(\lambda)$ and the reference values of $K_d^0(\lambda)$ provided by Jerlov’s measurements for each water type. The optimization parameters uniquely define absorption, scattering, and diffuse attenuation coefficients via Eqs. (5–9) and allow for the computation of average molecular scattering using Eq. (4). Below, a

Table 2. Key parameters of the BSA algorithm

Parameter	Value	Comment
Number of tries/run	3000	Maximum number
Cooling schedule	$0.96k$	k is the try number
Stop criteria	10^{-7}	Parameters’ delta
Number of re-runs	20–30	Max range Eq. (11)
Number of repeats	5–7	20–30 runs each

method to reduce the number of independent optimization parameters is suggested on the assumption that the generic water type is known (i.e., coastal versus oceanic). In this case, the number of parameters can be reduced from five to only three: Chl , B_s , and B_l .

4. COMPARISON WITH HYDROLIGHT

A check on the quality of the retrieved absorption and scattering coefficients was made by using the retrieved a and b spectra as inputs to the HydroLight radiative transfer numerical model [13]. Those IOPs, along with the assumption of a Petzold average-particle phase function and appropriate boundary conditions (sun zenith angle, clear sky, infinite water depth) allow HydroLight to compute the in-water radiance distribution. The corresponding irradiances and diffuse attenuation functions are standard outputs from HydroLight. The HydroLight model was run without inelastic scatter because original Jerlov data were taken at relatively small depths of few scattering lengths (<25 m typical), where inelastic scattering was not a substantial factor. That as well corresponds to the conditions of the BSA inversion. The HydroLight-computed K_d values, denoted K_d^H in the tables of Appendix A, are compared with the original Jerlov K_d^0 , in Fig. 4. This figure shows that the matches between the HydroLight-computed and the Jerlov K_d spectra are reasonably close. Figure 5 shows the percent differences for all curves, computed as $100(K_d^0 - K_d^H)/K_d^0$. Ninety percent of the plotted points are within $\pm 20\%$. The retrieved $a(\lambda)$ and $b(\lambda)$ spectra, along with a Petzold average-particle phase function, therefore provide a consistent set of IOPs for each Jerlov water type, to within an error mostly better than $\pm 20\%$ in the corresponding $K_d(\lambda)$ spectra.

5. RESULTS AND DISCUSSION

Recovered values of Chl (chlorophyll concentration), B_s and B_l (concentrations of small and large particles), and α and M are given in Table 3 for all Jerlov water types, along with standard deviations. Values of the recovered a , b , and K_d as functions of wavelength are provided in Appendix A. A representative example of reconstructed $a(\lambda)$ and $b(\lambda)$ is shown in Fig. 6 for Jerlov type 1C coastal water (Table 7). As seen from the tables in Appendix A, the values of K_d resulting from the inversion solution match the reference Jerlov values K_d^0 with accuracy better than 15% at all wavelengths.

The reconstructed concentrations of chlorophyll (Table 3) vary from 0.01 mg m^{-3} (very clear oceanic Jerlov type I water) to 9.1 mg m^{-3} (very turbid coastal or river-discharge water). Typical coastal water, Jerlov 1C, yields chlorophyll concentrations of about 1 mg m^{-3} , in line with data reported by Smart

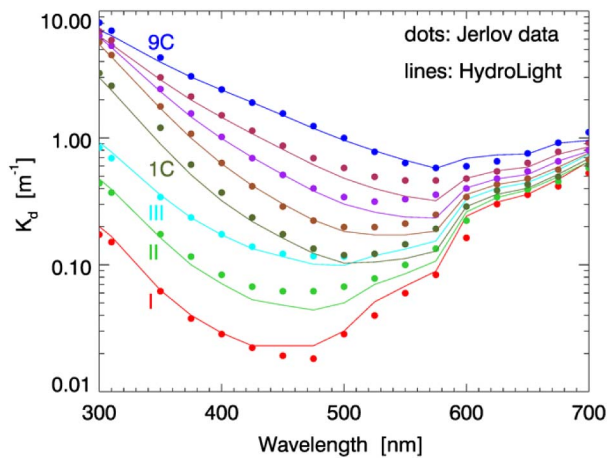


Fig. 4. K_d spectra computed by HydroLight (solid curves) using the absorption and scattering spectra retrieved by the BSA algorithm. The dots are the reference spectra. The color coding identifies the Jerlov water types.

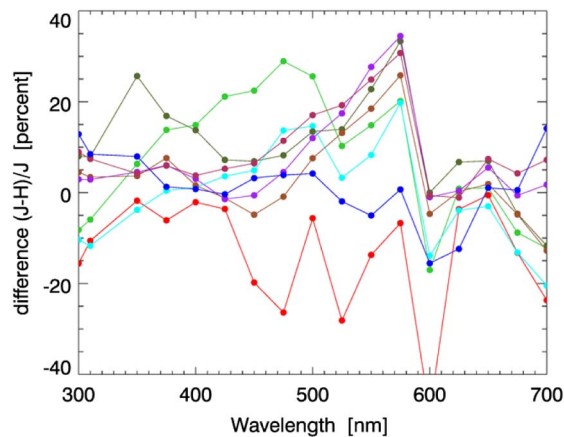


Fig. 5. Percent differences between the K_d spectra computed by HydroLight and the Jerlov reference spectra. Color coding is the same as in Fig. 4.

[7], as derived from NASA MODIS Aqua measurements. However, the standard procedure for reconstructing chlorophyll concentrations from MODIS Aqua data (see, for example, [27]) can sometimes overestimate chlorophyll concentrations for lower values ($< 1 \text{ mg m}^{-3}$) by as much as a factor of 3 [28,29]. This discrepancy occurs when large $K_d(490)$ is caused by scattering associated with higher concentrations of organic and inorganic particulates that are, however, smaller in size, thus resulting overall in lower chlorophyll concentration. This situation is well illustrated by comparing the IOPs and AOPs for Jerlov III (Table 6) and Jerlov 1C (Table 7) water types. Both water types have essentially the same values of K_d down to about 500 nm (Fig. 1), but at shorter wavelengths, attenuation increases significantly faster with decreasing wavelength in water type 1C. The IOP modeling reveals underlying differences in water content: Jerlov III water has a lower chlorophyll concentration than Jerlov 1C (0.177 versus 1.06 mg m^{-3}) but a higher concentration of small particulate matter. The result is lower absorption and higher scattering at 490 nm for Jerlov III and a similar $K_d(490)$ for both waters. Such considerations are important when deducing chlorophyll concentrations from remotely sensed ocean color data.

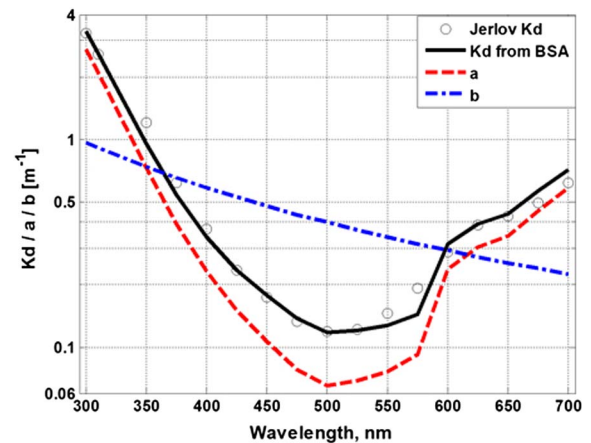


Fig. 6. BSA solution spectrum of K_d compared to the reference spectrum K_d^0 for Jerlov water type 1C (typical coastal water), and the corresponding solution IOPs $a(\lambda)$ and $b(\lambda)$.

Table 3. BSA Results for Chlorophyll Concentrations and Parameters Associated with Particulates and CDOM for Jerlov Water Types I–9C^a

Jerlov Water	Chl, mg m^{-3}	B_l , g m^{-3}	B_s , g m^{-3}	η	M	α , nm^{-1}
I	0.010 ($3 \cdot 10^{-4}$)	$2 \cdot 10^{-4}$ ($8 \cdot 10^{-5}$)	$8 \cdot 10^{-5}$ ($5 \cdot 10^{-5}$)	0.93 (0.02)	2.34 (0.06)	0.018 ($7 \cdot 10^{-4}$)
IA	0.027 ($4 \cdot 10^{-4}$)	0.005 ($2 \cdot 10^{-3}$)	0.002 ($1 \cdot 10^{-3}$)	0.44 (0.004)	1.69 (0.02)	0.020 ($3 \cdot 10^{-4}$)
IB	0.037 (0.001)	0.083 (0.012)	0.03 (0.006)	0.06 (0.003)	1.49 (0.06)	0.022 ($2 \cdot 10^{-4}$)
II	0.044 (0.001)	0.011 (0.005)	0.401 (0.007)	0.007 ($1 \cdot 10^{-3}$)	1.31 (0.02)	0.023 ($1 \cdot 10^{-4}$)
III	0.177 (0.01)	0.006 (0.004)	1.1 (0.02)	0.003 ($2 \cdot 10^{-4}$)	0.95 (0.03)	0.024 ($2 \cdot 10^{-4}$)
1C	1.00 (0.05)	0.004 (0.002)	0.402 (0.03)	0.005 ($4 \cdot 10^{-4}$)	1.22 (0.03)	0.027 ($4 \cdot 10^{-4}$)
3C	1.28 (0.07)	0.005 (0.002)	1.21 (0.05)	0.003 ($1 \cdot 10^{-4}$)	2.02 (0.07)	0.026 ($1 \cdot 10^{-4}$)
5C	3.95 (0.33)	0.022 (0.01)	1.50 (0.09)	0.001 ($1 \cdot 10^{-4}$)	1.89 (0.09)	0.022 ($5 \cdot 10^{-5}$)
7C	8.4 (1.2)	0.067 (0.02)	2.64 (0.5)	< 0.001	2.25 (0.33)	0.017 ($5 \cdot 10^{-4}$)
9C	9.1 (1.5)	0.016 (0.01)	3.54 (0.6)	< 0.001	4.32 (0.46)	0.015 ($1 \cdot 10^{-4}$)

^aStandard deviations are given in parentheses.

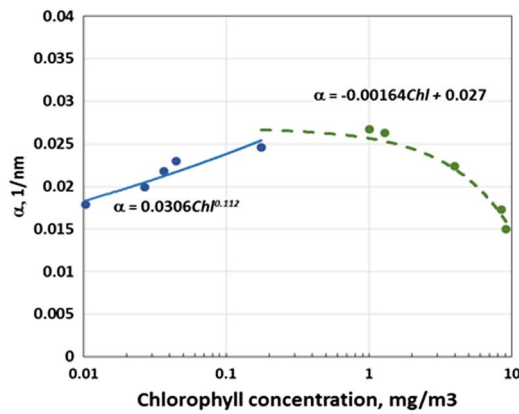


Fig. 7. Dependence of the CDOM exponential term α of Eq. (7) on chlorophyll concentration for oceanic (filled circles) and coastal (open circles) waters.

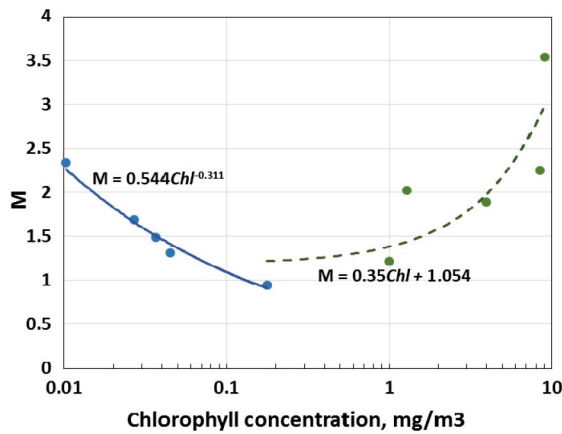


Fig. 8. Dependence of the CDOM multiplier M of Eq. (7) on chlorophyll concentration for oceanic (filled circles) and coastal (open circles) water types.

This work found that the terms α and M of Eq. (7) both depend on chlorophyll concentration and, more importantly, on generic water type (i.e., oceanic versus coastal). These dependencies, together with relevant fits, are presented in Figs. 7 and 8.

The Jerlov I water data were not used in the development of the fits shown in Figs. 7 and 8, because the CDOM and chlorophyll concentrations for this water type were so low that for wavelengths above ~ 450 nm the K_d values were driven largely by pure-water absorption. As a result, the search algorithm exhibited poor sensitivity to variations in chlorophyll and CDOM concentrations, resulting in large optimization-related errors. To regularize the BSA for Jerlov I, a scattering coefficient of pure water at 300 nm was taken from [23] and incorporated into the cost function, Eq. (10), as $\delta = (b_{\text{ref}} - b_{\text{BSA}})/b_{\text{ref}}$, where b_{ref} is the reference value of the total scattering coefficient of purest ocean water at 300 nm (0.0262 m^{-1}). Then, the fit parameters developed for Jerlov waters IA through 9C were used to obtain particulate and chlorophyll concentrations for Jerlov I water.

Incorporation of the fitting results (Figs. 7 and 8) into Eq. (7) gives expressions for CDOM absorption in ocean and coastal waters in terms of chlorophyll concentration and wavelength. Using values of $A(440) = 0.052$ and $E(440) = 0.635$ [14] gives

$$a_{\text{cdom}}^o(\lambda) = 0.0235[\text{Chl}]^{-p} \exp[-0.029(\lambda - 440)],$$

$$p = 0.00235(\lambda - 440) - 0.316, \quad (12)$$

$$a_{\text{cdom}}^c(\lambda) = (0.0102 \text{ Chl} + 0.0614)[\text{Chl}]^{0.635} \exp[-\beta(\lambda - 440)],$$

$$\beta = -0.00133 \text{ Chl} + 0.028. \quad (13)$$

Thus, expressing CDOM absorption in terms of chlorophyll concentration reduces the dimensionality of the BSA fit to only 3. This reduction of the number of independent parameters results in smaller algorithm-related errors and enables the use of less computationally intensive optimization

Table 4. Jerlov I and IA Waters

Wavelength, nm	Jerlov I					Jerlov IA				
	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}
300	0.173	0.186	0.200	0.163	2.08E-02	0.223	0.241	0.266	0.221	2.55E-02
310	0.151	0.154	0.167	0.134	1.81E-02	0.186	0.199	0.218	0.181	2.26E-02
350	0.0619	0.059	0.063	0.048	1.08E-02	0.078	0.0776	0.085	0.0673	1.45E-02
375	0.0377	0.038	0.040	0.030	8.11E-03	0.050	0.0490	0.053	0.0413	1.14E-02
400	0.0284	0.028	0.029	0.022	6.20E-03	0.0377	0.0356	0.038	0.0295	9.20E-03
425	0.0222	0.022	0.023	0.017	4.82E-03	0.0294	0.0274	0.029	0.0225	7.55E-03
450	0.0192	0.022	0.023	0.018	3.81E-03	0.0263	0.0264	0.027	0.0221	6.31E-03
475	0.0182	0.021	0.023	0.019	3.06E-03	0.0253	0.0253	0.026	0.0216	5.36E-03
500	0.0284	0.029	0.030	0.026	2.49E-03	0.0346	0.0316	0.033	0.0282	4.61E-03
525	0.0398	0.049	0.051	0.046	2.05E-03	0.0460	0.0503	0.052	0.0468	4.02E-03
550	0.0598	0.065	0.068	0.062	1.70E-03	0.0661	0.0658	0.068	0.0622	3.54E-03
575	0.0834	0.085	0.089	0.082	1.43E-03	0.0943	0.0858	0.089	0.0821	3.15E-03
600	0.163	0.233	0.242	0.228	1.22E-03	0.174	0.234	0.242	0.228	2.83E-03
625	0.301	0.302	0.312	0.295	1.04E-03	0.308	0.303	0.312	0.295	2.56E-03
650	0.357	0.341	0.359	0.334	8.99E-04	0.364	0.342	0.359	0.334	2.34E-03
675	0.416	0.444	0.471	0.434	7.82E-04	0.423	0.445	0.471	0.435	2.14E-03
700	0.528	0.595	0.653	0.582	6.85E-04	0.536	0.595	0.653	0.582	1.98E-03

techniques, such as the Levenberg–Marquardt or orthogonal descent methods.

Values of the exponential parameter α in Eq. (7) obtained in this analysis for coastal-water CDOM ($\alpha = 0.015$ to 0.027) are within the range of measured values (0.012 to over 0.035 [30]). The value of the multiplier M derived here is substantially larger than the one used previously (1 to 4 in this work versus 0.2 in [14]). This difference may be due to the different relations between CDOM and chlorophyll concentrations in the Jerlov waters versus the relations implied by the Case 1 water data in [14]. As noted in [31], CDOM absorption of turbid (C5 to C9) Jerlov waters is particular to the Baltic Sea and certain coastal waters where Jerlov measurements were taken. As seen in Fig. 7, M and chlorophyll are, not surprisingly, less well correlated for coastal water types than for oceanic water types (correlation coefficient $r^2 = 0.91$ versus 0.98). The generally

larger M values and the lower correlation coefficient for coastal waters reflect the fact that absorption associated with terrestrial runoff does not correlate well with chlorophyll concentration, and the adopted value of the average scattering cosine ($g = 0.924$) does not remain constant for different water types. These conditions result in some additional errors in retrieved IOPs for waters of Jerlov types 3C, 5C, 7C, and 9C in particular.

Table 3 shows that the recovered chlorophyll concentrations increase with progression from clear to turbid water types. For type I clear oceanic waters, scattering is caused mostly by larger particles ($B_l > B_s$). For Jerlov types II through 5C, scattering losses are mostly related to the presence of small particles. This is due to the particular choice of a small particle scatter function [Eq. (8c)] that has a greater short wavelength scatter than a large particle scatter function [Eq. (8d)]. For more turbid waters, the

Table 5. Jerlov IB and II Waters

Wavelength, nm	Jerlov IB					Jerlov II				
	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}
300	0.288	0.301	0.333	0.273	0.125	0.439	0.464	0.475	0.343	1.014
310	0.223	0.246	0.271	0.221	0.118	0.371	0.384	0.393	0.273	0.957
350	0.0998	0.0986	0.106	0.0810	0.0968	0.174	0.171	0.163	0.095	0.776
375	0.0619	0.0629	0.065	0.0481	0.0872	0.116	0.114	0.100	0.0540	0.689
400	0.0460	0.0460	0.046	0.0331	0.0795	0.0834	0.0846	0.071	0.0355	0.616
425	0.0367	0.0360	0.035	0.0246	0.0732	0.0672	0.0671	0.053	0.0257	0.555
450	0.0336	0.0342	0.032	0.0235	0.0680	0.0619	0.0620	0.048	0.0241	0.504
475	0.0325	0.0325	0.030	0.0225	0.0635	0.0619	0.0579	0.044	0.0228	0.459
500	0.0408	0.0386	0.036	0.0287	0.0597	0.0672	0.0641	0.050	0.0288	0.421
525	0.0513	0.0572	0.056	0.0469	0.0565	0.0780	0.0845	0.070	0.0469	0.387
550	0.0726	0.0726	0.072	0.0623	0.0536	0.0998	0.0999	0.085	0.0622	0.358
575	0.0998	0.0926	0.093	0.0822	0.0511	0.134	0.120	0.107	0.0821	0.332
600	0.186	0.241	0.248	0.2278	0.0488	0.223	0.270	0.261	0.228	0.309
625	0.315	0.310	0.319	0.296	0.0468	0.342	0.337	0.339	0.296	0.288
650	0.371	0.349	0.364	0.334	0.0450	0.393	0.375	0.389	0.334	0.270
675	0.431	0.452	0.475	0.435	0.0434	0.454	0.476	0.494	0.436	0.253
700	0.545	0.602	0.653	0.582	0.0420	0.580	0.624	0.651	0.582	0.238

Table 6. Jerlov III and Jerlov IC Waters

Wavelength, nm	Jerlov III					Jerlov IC				
	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}
300	0.837	0.860	0.924	0.568	2.76	(3.244)	3.294	2.984	2.686	1.037
310	0.693	0.719	0.774	0.452	2.61	(2.577)	2.576	2.373	2.083	0.979
350	0.342	0.343	0.355	0.164	2.12	1.204	0.947	0.895	0.721	0.793
375	0.236	0.234	0.235	0.094	1.88	0.616	0.539	0.512	0.386	0.704
400	0.174	0.175	0.172	0.0615	1.69	0.371	0.340	0.320	0.227	0.630
425	0.139	0.140	0.134	0.0449	1.52	0.236	0.236	0.219	0.147	0.567
450	0.122	0.124	0.116	0.0388	1.38	0.174	0.179	0.162	0.105	0.514
475	0.117	0.110	0.101	0.0335	1.26	0.134	0.139	0.123	0.077	0.469
500	0.116	0.109	0.099	0.0358	1.152	0.119	0.120	0.103	0.064	0.429
525	0.122	0.129	0.118	0.0507	1.06	0.122	0.121	0.105	0.068	0.395
550	0.145	0.144	0.133	0.0646	0.980	0.145	0.129	0.112	0.076	0.365
575	0.192	0.164	0.154	0.0838	0.908	0.192	0.146	0.128	0.092	0.338
600	0.288	0.324	0.328	0.229	0.845	0.288	0.317	0.288	0.236	0.314
625	0.386	0.390	0.401	0.297	0.788	0.386	0.395	0.360	0.304	0.293
650	0.431	0.426	0.444	0.336	0.737	0.431	0.439	0.401	0.344	0.274
675	0.494	0.528	0.559	0.439	0.692	0.494	0.568	0.517	0.455	0.257
700	0.616	0.672	0.742	0.583	0.650	0.616	0.721	0.688	0.586	0.242

algorithm increases the concentration of smaller particles in order to account for the increase in short wavelength absorption better. For very turbid waters (types 7C and 9C), the BSA output for small versus large particle concentrations was statistically ambiguous. For Jerlov 7C and Jerlov 9C waters, the information content of the field data (specifically, the number of data points and the measurement accuracy) is poor, resulting in significant fluctuations of B_l and B_s as the minimized cost function varies insignificantly. Therefore, values of B_l and B_s for Jerlov 7C and 9C have poor fidelity (large standard deviation).

6. CONCLUSIONS

Various bio-optical models relate absorption $a(\lambda)$ and scattering $b(\lambda)$ coefficients to the diffuse attenuation $K_d(\lambda)$, and to

the chlorophyll concentration in Case 1 waters. We combined such models with an inversion algorithm based on BSA to retrieve absorption and scattering coefficients that are consistent with the K_d spectra used to define the Jerlov water types. The solutions obtained via the BSA algorithm used a cost function [Eq. (10)] that depends on the absolute difference between the $K_d(\lambda)$ computed from the bio-optical models using the a and b values at the current iteration and the K_d^0 spectra used to define the Jerlov water types. The solutions for a and b values (the solution IOPs) are those that minimize the cost function. When used in the bio-optical models, the solution IOPs give $K_d(\lambda)$ values that are within 15% of the Jerlov values (Fig. 6 and Appendix A). The solution IOPs, when used in HydroLight, give $K_d(\lambda)$ spectra that are within $\pm 20\%$ of the Jerlov values 90% of the time (Figs. 4 and 5). The solution IOPs also relate the Jerlov water types to the chlorophyll

Table 7. Jerlov 3C and 5C Waters

Wavelength, nm	Jerlov 3C					Jerlov 5C				
	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}
300	(5.705)	5.934	5.449	4.733	3.00	(6.42)	6.75	6.232	5.36	3.73
310	(4.507)	4.659	4.353	3.668	2.83	(5.33)	5.53	5.176	4.34	3.53
350	1.772	1.781	1.707	1.287	2.30	2.43	2.42	2.320	1.78	2.87
375	1.079	1.032	0.997	0.685	2.04	1.56	1.52	1.466	1.05	2.55
400	0.635	0.648	0.625	0.388	1.83	1.02	1.02	0.989	0.660	2.28
425	0.416	0.439	0.421	0.236	1.65	0.693	0.729	0.703	0.437	2.06
450	0.288	0.319	0.302	0.154	1.50	0.511	0.535	0.514	0.297	1.87
475	0.223	0.240	0.225	0.105	1.36	0.400	0.401	0.382	0.204	1.71
500	0.198	0.197	0.183	0.081	1.25	0.342	0.319	0.301	0.151	1.56
525	0.198	0.187	0.172	0.078	1.15	0.315	0.277	0.260	0.127	1.44
550	0.211	0.187	0.172	0.082	1.06	0.329	0.255	0.238	0.117	1.33
575	0.248	0.201	0.184	0.095	0.985	0.357	0.252	0.234	0.119	1.23
600	0.342	0.380	0.358	0.239	0.916	(6.42)	6.75	6.232	5.36	3.73
625	0.431	0.456	0.430	0.307	0.855	(5.33)	5.53	5.176	4.34	3.53
650	0.478	0.498	0.469	0.346	0.800	2.43	2.42	2.320	1.78	2.87
675	(5.705)	5.934	5.449	4.733	3.00	1.56	1.52	1.466	1.05	2.55
700	(4.507)	4.659	4.353	3.668	2.83	1.02	1.02	0.989	0.660	2.28

Table 8. Jerlov 7C and Jerlov 9C Waters

Wavelength, nm	Jerlov 7C					Jerlov 9C				
	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}	K_d^0, m^{-1}	K_d, m^{-1}	K_d^H, m^{-1}	a, m^{-1}	b, m^{-1}
300	(6.92)	6.67	6.303	5.11	6.56	(8.11)	7.44	7.066	5.466	8.76
310	(5.91)	5.80	5.471	4.40	6.20	(7.01)	6.72	6.416	4.900	8.28
350	3.00	3.05	2.874	2.17	5.04	4.30	4.11	3.958	2.856	6.73
375	2.12	2.13	1.994	1.45	4.49	3.06	3.13	3.021	2.103	5.99
400	1.51	1.59	1.453	1.03	4.02	2.41	2.48	2.391	1.613	5.36
425	1.14	1.21	1.080	0.753	3.63	1.90	1.97	1.907	1.242	4.84
450	0.868	0.924	0.812	0.542	3.30	1.56	1.56	1.510	0.943	4.39
475	0.693	0.710	0.614	0.388	3.01	1.24	1.24	1.192	0.709	4.01
500	0.580	0.560	0.481	0.290	2.76	1.00	1.00	0.958	0.543	3.67
525	0.494	0.470	0.399	0.233	2.54	0.777	0.826	0.792	0.430	3.38
550	0.462	0.409	0.347	0.195	2.35	0.635	0.696	0.667	0.348	3.12
575	0.462	0.371	0.320	0.175	2.18	0.580	0.603	0.576	0.291	2.89
600	0.478	0.527	0.481	0.301	2.03	0.598	0.720	0.691	0.390	2.69
625	0.545	0.600	0.551	0.367	1.89	0.654	0.763	0.735	0.436	2.51
650	0.635	0.635	0.588	0.403	1.77	0.755	0.775	0.747	0.456	2.35
675	0.777	0.815	0.744	0.559	1.66	0.916	0.945	0.911	0.604	2.20
700	0.916	0.874	0.850	0.621	1.56	1.11	0.990	0.953	0.651	2.07

concentration. A self-consistent set of IOPs, K_d , and Jerlov water type were thereby obtained. This association of IOPs and chlorophyll concentration with Jerlov water type enhances the usefulness of the Jerlov classification scheme, which is of potential importance for the underwater visibility, communication, and remote sensing communities and provides insight into the accuracy of common IOP spectral models.

APPENDIX A

Tables 4–8 give, for the different Jerlov water types, the reference-value diffuse attenuation coefficients $K_d^0(\lambda)$ from Jerlov's data, the reconstructed diffuse attenuation coefficients $K_d(\lambda)$ from the bio-optical IOP models and the BSA algorithm, the diffuse attenuation coefficients $K_d^H(\lambda)$ obtained from the HydroLight model using the recovered absorption and scattering coefficients, and the absorption and scattering coefficients themselves. Values in parentheses were obtained by extrapolation. For all but Jerlov III waters the HydroLight modeling depth was set to $bz = 10$ scattering lengths, while for Jerlov III water a better match between Jerlov and HydroLight was achieved for $bz = 6$.

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